Blizzard Impact on School Attendance

CSD 4 (East Harlem) | Grades K-4 | Prediction for February 24, 2026

Context

On February 22, 2026, the Blizzard of 2026 dropped 22 inches of snow on NYC. Schools closed Monday the 23rd and reopened Tuesday the 24th with MTA service running at reduced capacity. CSD 4 serves 3,196 K-4 students across 18 elementary schools in East Harlem, where 86% of students are economically disadvantaged. This analysis uses 84 days of attendance and weather data to predict how many students would attend on the return day.

Key Findings

75.0%	2,397	~415	72-82%
Predicted Attendance	Students Expected	Extra Absences	95% Confidence Interval

Hypothesis Testing: Prediction vs Actual

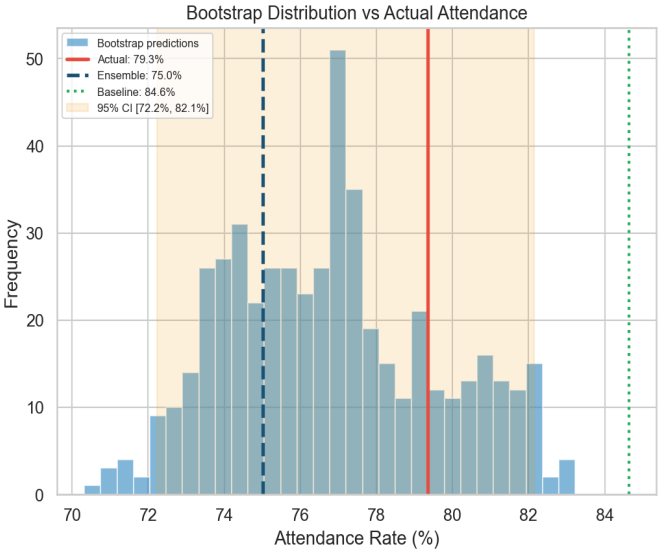
Actual Feb 24 attendance was 79.3% (2,536 students). The model predicted 75.0%. The actual value falls within the 95% confidence interval [72.2%, 82.1%], confirming the model was well-calibrated.

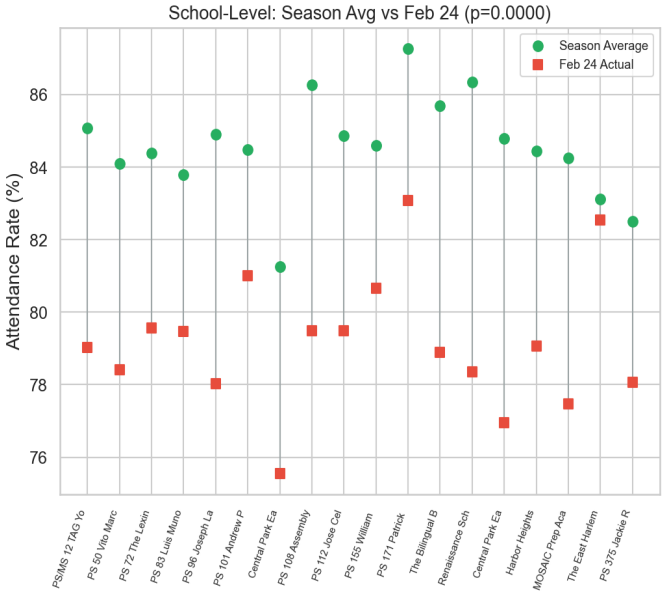
Scorecard

Blizzard effect is real?	**YES**	*Paired t-test p < 0.0001*
Effect size is meaningful?	**YES**	*Cohen's d = -3.11 (large)*
Prediction within 95% CI?	**YES**	*Actual 79.3% in [72.2%, 82.1%]*
Model statistically accurate?	**YES**	*Bootstrap z-test p = 0.35*

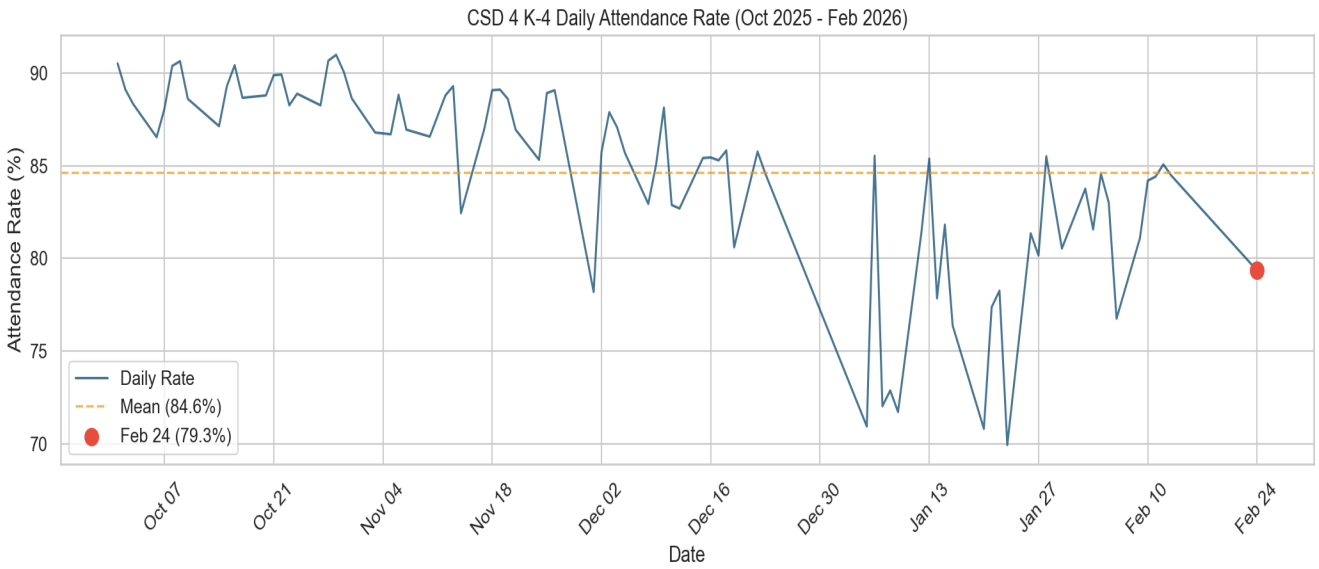
Verdict: STRONG SUPPORT - the blizzard effect is real, and the model is operationally useful.

Prediction vs Actual





Attendance Over Time



Recommendations

1. Pre-position tutoring and program staff based on 70-80% attendance forecast before reopening.
2. Target outreach to chronically absent families the evening before return day.
3. Delay assessments by one day to avoid mass makeups and scheduling disruption.
4. Share attendance forecasts with meal programs so they can adjust food prep quantities.
5. Run this model weekly using weather forecasts to flag any sub-80% attendance days in advance.

Methodology: Weighted ensemble of OLS regression, Random Forest (n=200), and Gradient Boosting (n=200) trained on 83 school days of attendance and NOAA weather data. 95% CI via 500-iteration bootstrap. Data sources: NYC DOE attendance records, NYSED 2023-24 enrollment, NOAA Central Park observations.